Coding Agent Memory Post-Training II: Unlocking the Memory Potential of Pre-trained File Operations for Long-Horizon Tasks via On-Policy Distillation

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Abstract

Language-model agents increasingly tackle long-horizon tasks whose interaction histories exceed the model's active context. They must therefore decide what to write down and what to read back, and when to do either. Recent work has begun to use on-policy distillation (OPD) to make memory control part of the policy, relying on memory tools and compaction protocols that lie outside either model's pre-training. However, this weakens the distillation on both sides: the teacher cannot give useful guidance on an interface it has never used, and the student samples too few meaningful memory actions for that guidance to take effect. To study this problem, we introduce Coding Agent Memory Gym II (CAMG-II), a suite of long-horizon agentic-RL environments spanning Shop, Coding, AutoResearch, General, and Visual. Alongside each environment's native task interface, CAMG-II provides ordinary Bash access and an episode-persistent workspace, enabling the policy to create, revise, search, and reuse files as memory across context replacements. We also introduce Coding Agent Memory Post-Training II (CAMP-II), which trains a single policy jointly across all five environments with fully asynchronous OPD to learn this file-based memory behavior directly from a frozen teacher with a prior for file operations. On CAMG-II, CAMP-II obtains 2.97 times the gain of the strongest OPD baseline. The memory behavior CAMP-II learns generalizes to two external benchmarks, Terminal-Bench 2.1 and SWE-Bench Pro V2, with improved success rates.

1. Introduction

Language-model agents increasingly tackle long-horizon tasks that unfold through repeated tool use and environment interaction [19, 65]. As these interactions accumulate, information needed for later decisions can fall out of the model's active context. A website-building agent, for example, may need to recall why an earlier visual correction was rejected after that exchange has left the prompt (Fig. 1).

Recent work applies on-policy distillation (OPD) to memory control through dedicated memory tools and compaction protocols [10, 27, 28, 34, 79]. However, this weakens the distillation on both sides: the teacher cannot give useful guidance on an interface it has never used, and the student samples too few meaningful memory actions for that guidance to take effect [13]. Moreover, unfamiliarity with the tool produces the negative-advantage problem: the teacher's scores push the student away from the memory behavior being taught without indicating the behavior that should take its place, which leaves training unstable [16, 80]. This raises a question: can a teacher teach a smaller model to control memory through operations that both the teacher and the student already have a prior for?

To study this question, we introduce **Coding Agent Memory Gym II (CAMG-II)**, a suite of long-horizon agentic-RL environments spanning Shop, Coding, AutoResearch, General, and Visual. Across all five environments, CAMG-II pairs each native task interface with ordinary Bash access and an episode-persistent workspace. Agents can therefore create, revise, search, and reuse files as memory throughout an episode.

We also introduce **Coding Agent Memory Post-Training II (CAMP-II)**, which trains a single policy jointly across all five environments with fully asynchronous OPD to learn this file-based memory behavior directly from a frozen teacher with a prior for file operations. Task-reward training of file-based memory asks one question, whether the student's pre-training prior makes the behavior reachable by exploration. Distillation asks two. The student has to emit meaningful memory operations, and the frozen teacher has to judge what those operations write. A shared interface prior answers both at once. We train CAMP-II-4B and CAMP-II-9B from Qwen3.5 models of matching size. On CAMG-II, CAMP-II obtains 2.97 times the gain of the strongest OPD baseline, and the memory behavior it learns generalizes to two external benchmarks, Terminal-Bench 2.1 and SWE-Bench Pro V2, with improved success rates.

Our contributions are threefold.

- **A two-sided interface prior for memory distillation.** We identify what an interface has to provide for OPD to teach memory at all, coverage of the action on the student side and supervision quality on the teacher side, and we show that a shared file-operation prior supplies both.

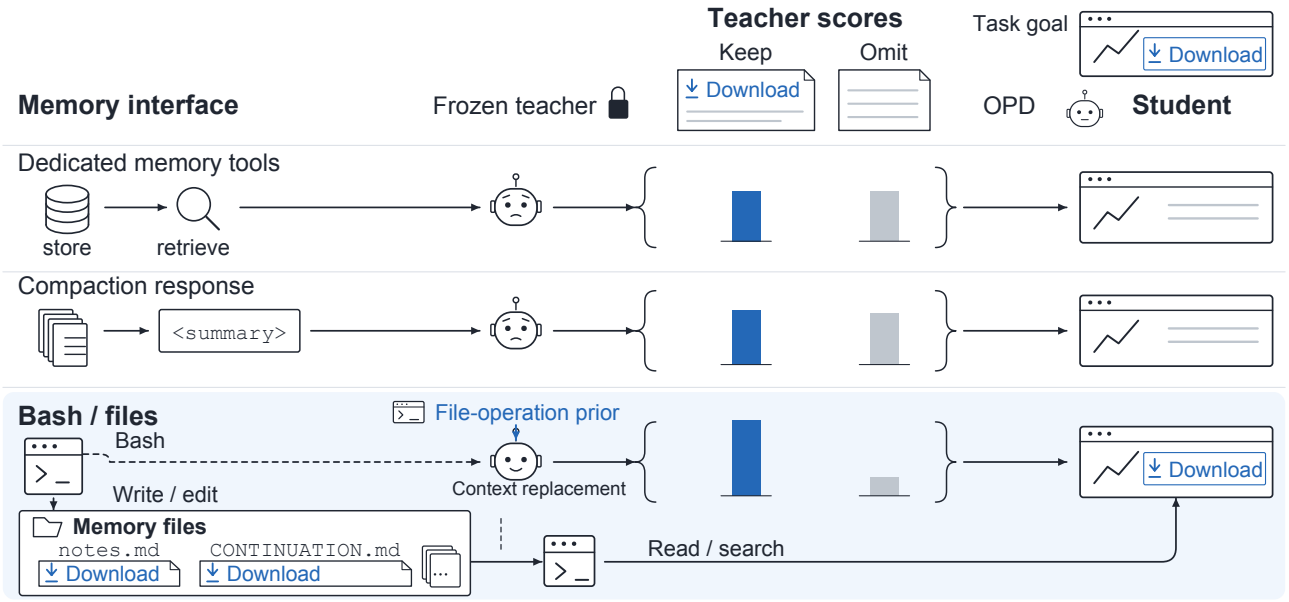


Figure 1. **File-operation priors improve supervision for memory distillation.** A frozen teacher scores student-generated memory responses that retain (*Keep*) or omit (*Omit*) the same Download requirement. Compared with dedicated memory tools and context-compaction protocols, ordinary Bash/file operations draw on the teacher's existing prior, yielding a clearer preference for retaining task-critical information and thus more effective supervision. The student's general notes and bounded continuation file persist across context replacements, so it can recover the saved Download requirement through Bash.

- **An on-policy distillation method that uses it.** We introduce CAMP-II, which learns this file-based memory behavior by on-policy distillation from a teacher that needs no memory-specific adaptation, and reaches the highest average success rate across the five test environments.
- **A long-horizon agentic-RL environment suite.** We introduce CAMG-II with five environments that require memory across context boundaries, each pairing native task interfaces with ordinary Bash access and an episode-persistent workspace.

2. Related Work

Agentic RL for Memory. These methods use reinforcement learning to improve how agents retain and recover task information. Existing work optimizes state consolidation and summary retention [26, 86], while other approaches train agents to use dedicated memory operations [79]. Context compaction is also learned from downstream outcomes [28], and agents are trained to manage their own context proactively through folding [75]. Other work uses task rewards to train agents to maintain state through ordinary filesystem actions, including bounded continuation-file compression [37]. These methods spend much of their exploration on invalid calls and failed executions [10], which makes a denser target attractive. CAMP-II also trains the policy, but the signal is token-level teacher targets rather than task reward.

Memory Systems for LLM Agents. These systems preserve interaction history under a bounded active context through different representations and memory-management architectures. Deployed frameworks give an agent persistent memory outside its window: a long-term store maintained by an extraction and update pipeline [6], hierarchical memory edited under an operating-system abstraction [42], a note collection that links and revises itself as the task proceeds [68], and an interaction history organized as a temporal knowledge graph [47]. All four are inference-time layers: the agent queries a store that sits outside the weights, and the task policy itself is never trained for the memory behavior. CAMP-II trains the task policy itself, and its memory is kept in ordinary files that the teacher can read as well.

On-Policy Distillation for Agent Memory. OPD supervises student-generated prefixes with teacher predictions [1]. For memory-dependent execution, existing approaches reconstruct the compact-memory invocation state the student acted in [34], annotate trajectories before distilling [27], or train a separate memory controller or writer through OPD followed by RL [10, 30]. Each aligns the two models on the state or on the trajectory and leaves the memory interface as given. That interface is a compact-

invocation protocol or a learned writer that neither model has practiced. CAMP-II instead selects the interface from the prior the teacher and the student share, and supervises memory content through it from a general teacher without memory-specific adaptation.

3. Preliminaries

Multi-turn interaction. We consider finite-horizon, partially observable tasks. Given the instruction and visible context h_t, the student samples response a_t. The environment executes the response and returns an observation for the next invocation. A trajectory τ contains $T_\tau \leq H$ invocations, where H is the horizon. Response a_t contains L_t tokens, and $a_{t,<j}$ is its prefix before token j. Write $s_{t,j} = (h_t, a_{t,<j})$ for the context of that token prediction.

On-policy distillation. OPD trains on contexts and prefixes reached by the student [1]. For a sampled token x at prefix s, let $p_\theta(x \mid s)$ and $q(x \mid s)$ be its probabilities under the current student and frozen teacher. Sampling from the student makes the reverse-KL gradient a policy gradient whose per-token signal is the teacher–student log-ratio,

$$A_\theta^{\mathrm{OPD}}(s,x) = \mathrm{sg}[\log q(x \mid s) - \log p_\theta(x \mid s)]\,,$$
$$\mathcal{L}_{\mathrm{PG}}(\theta) = -\mathbb{E}_{(s,x)\sim\pi_\theta}\left[A_\theta^{\mathrm{OPD}}(s,x)\,\log p_\theta(x \mid s)\right]. \tag{1}$$

Here sg stops gradients through its argument.

4. Coding Agent Memory Gym II

To train agents to use files as memory, we construct CAMG-II as a suite of five long-horizon environments (Fig. 2) that extends CAMG's environments [37] with enterprise knowledge work and visually grounded website development, drops DeepResearch, an environment that is heavy to run and hard to optimize with reinforcement learning, and refines the reward design and the tool format. Each environment combines verifiable tasks with ordinary Bash access and an episode-persistent workspace. Their tasks require agents to reuse information accumulated earlier in the episode.

Every environment shares one interface. The workspace persists for the whole episode and the active context does not: when the context approaches its budget, it is cleared, so anything the next steps need has to have been written down. Before that replacement the policy writes a bounded continuation file carrying the task state, the replacement proceeds once that write is valid, and the context that follows contains only that file's path. The state therefore survives a replacement when the policy writes it and reads it back afterwards.

4.1. Constructing Five Long-Horizon Environments

Shop. Shop tests whether an agent can retain customer requirements and earlier purchase decisions across a sequence of dependent purchases. We generate six-session tasks from 720 verified WebShop products [74]. Between sessions the environment clears the active session transcript, so later purchases depend on what the policy retains.

Coding. Coding tests whether an agent can complete a software repair whose diagnosis, repository edits, and test evidence may span multiple active contexts. We draw issue–repository pairs and test-defined repair objectives from SWE-smith [70]. We exclude non-executable tasks and those whose reference repairs fail their tests, retaining 11,218 tasks from 93 repositories.

AutoResearch. AutoResearch tests whether an agent can preserve hypotheses and validation evidence across iterative model development. We use public OpenMLE task packages with datasets, executable submissions, and native metrics [71]. We exclude tasks that cannot run or whose reference pipelines fail, retaining 931 tasks from 804 task families.

General. General tests whether an agent can reconcile and retain evidence across business documents and system interactions. We adapt MiMo's enterprise tasks [66], with rubric-based judging of the answer, deliverables, and applicable system state. The adaptation also fixes incomplete tool signatures by exposing all declared arguments and distinguishing required from optional inputs.

Visual. Visual tests whether an agent can retain design requirements and visual inspection findings during website construction. We adapt MiMo's website briefs and visual grading [66]. The vision-capable agent builds a static site, inspects rendered previews, and revises the delivery. The delivered site receives an absolute rubric score for visual quality, brief fulfillment, and asset quality.

4.2. A Shared File-Memory Interface

Across the five environments, CAMG-II standardizes an executable filesystem interface for using files as memory while retaining each environment's native task interaction and outcome evaluation. Following the minimal, shell-only interface of mini-swe-agent [69], the shared filesystem action is a single `bash` entry, which can create, inspect, search, modify, and execute files, so the policy can preserve and revise task-relevant evidence and intermediate state in the workspace. Let $\mathcal{A}_d^{\mathrm{task}}$ denote environment d's native

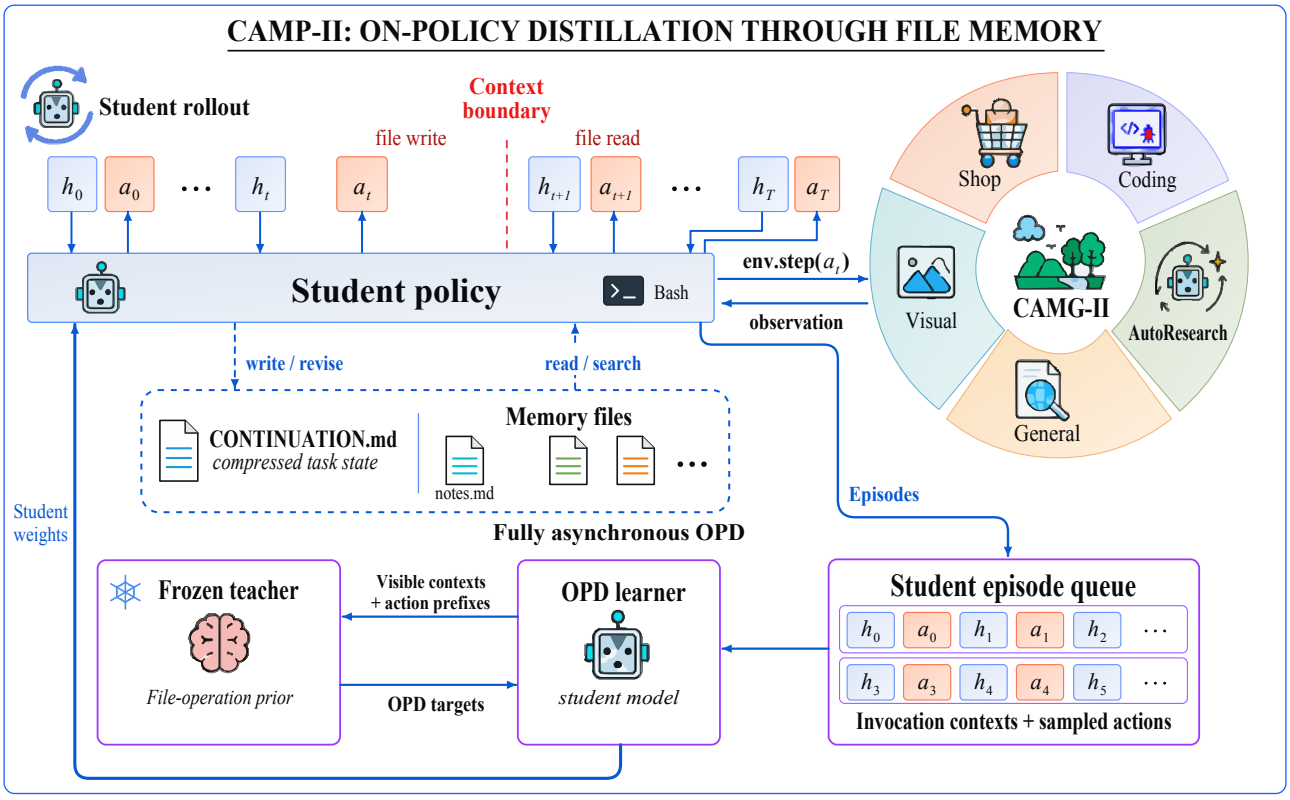


Figure 2. **CAMP-II distils file-based memory across five environments. Top:** the student acts in the environment and writes and revises memory files. At each context boundary it compresses its state into the continuation file, which a later read restores. **Bottom:** completed episodes enter the queue, a frozen teacher scores the tokens the student actually sampled, and the OPD learner updates the student weights from the teacher–student log-probability gap, closing the loop back to the rollout.

task-action set and $\mathcal{A}^{\text{fs}}$ the shared filesystem-action set. The CAMG-II action space is

$$\mathcal{A}_d = \mathcal{A}_d^{\text{task}} \cup \mathcal{A}^{\text{fs}}, \qquad a_t \in \mathcal{A}_d. \tag{2}$$

One `bash` entry can produce both native-task and filesystem effects, so the two sets may overlap. Appendix A.2 specifies the common interface. Figure 2 summarizes how this common interface connects task interaction and file use to the learning pipeline described next.

5. CAMP-II: File-Memory Distillation

Section 4 gives every task an episode-persistent workspace and a single shared `bash` action. This section uses that workspace to distill memory for long-horizon tasks, so that task state survives a context boundary. Figure 2 traces one turn: at turn t the student samples a response $a_t \sim \pi_\theta(\cdot \mid h_t)$ from the visible context h_t, with $a_t \in \mathcal{A}_d$ as defined in Eq. 2, and the teacher scores every token the student sampled, including the tokens of the file write that carries memory. Let i run over sampled response tokens across invocations and episodes, with x_i the token and s_i its recorded prefix, and write $p_{\theta,i} = \pi_\theta(x_i \mid s_i)$, $q_i = \pi_T(x_i \mid s_i)$, and $\mu_i = \pi_{\theta^-}(x_i \mid s_i)$ for the current student, the frozen teacher, and the same student at the weights that sampled the token. A write is part of a_t, so its tokens carry the same supervision as task tokens. After a context replacement only the path of the file the student wrote earlier is supplied, and the student reads it back to recover that state. The per-token signal is

$$A_i^{\text{OPD}} = \text{sg}[\log q_i - \log p_{\theta,i}]\,. \tag{3}$$

The rollout workers generate with the last published weights θ^- while the learner updates θ, so episodes from all five environments reach the learner after later weights are published. The update is corrected by the importance ratio $\rho_i = p_{\theta,i}/\mu_i$, taken against the clipped value $\bar{\rho}_i = \text{clip}(\rho_i, 1-\epsilon, 1+\epsilon)$. Over the sampled response tokens $\mathcal{B}^m$

of an optimizer minibatch, the objective is

$$\mathcal{L}^{m}_{\mathrm{OPD}}(\theta) = -\frac{1}{|\mathcal{B}^{m}|} \sum_{i \in \mathcal{B}^{m}} \min\left\{\rho_i A_i^{\mathrm{OPD}},\ \bar{\rho}_i A_i^{\mathrm{OPD}}\right\}, \quad (4)$$

where $\mathcal{B}^m$ excludes instructions, observations and padding, and m indexes the memory interface in the controlled comparisons. Equation 4 is the clipped form of the surrogate in Eq. 1. Algorithm 1 in Appendix C gives the procedure the workers and the learner run.

The prefix s_i is taken from the invocation the student actually produced: its visible context, token positions and causal mask [34], including invocations after a context replacement, together with the read observations the student recorded.

6. Experiments

We first compare CAMP-II with baselines on the five-domain test set and explain why file memory makes effective use of pre-training. We then measure external transfer and ablate the content of a write. Finally, we isolate the causal role of saved memory and follow one write from record to readback.

6.1. Experimental Setup

Training. All trainable methods use fresh Qwen3.5-4B and Qwen3.5-9B students [46] for 100 learner optimizer updates with 64 episodes per update, totaling 6,400 episodes per training run, under the equal mixture over Shop, Coding, AutoResearch, General, and Visual, trained with fully asynchronous OPD and no task rewards. Appendix C gives the remaining training details.

Baselines. We compare against four trainable memory-learning systems: compact memory with invocation reconstruction for MemOPD [34], context editing and retrieval of offloaded history for ACM [27], a hierarchical memory with provenance and validity checks for HELM [30], and a separate writer with a persistent bank for Hy-Memory [10]. For HELM we report both the published architecture, which pairs a learned memory controller with a separate task reasoner, and a single-model adaptation in which one policy fills both roles. For MemOPD we report the recipe without and with the paper's SFT cold start. As the task-reward reference, we additionally retrain the predecessor CAMP-I [37], which keeps the same file memory and learns from task rewards instead of the teacher's supervision. Every system except CAMP-I is trained by common reverse-KL OPD from the shared frozen Qwen3.8-27B teacher, so the systems differ only in how they keep memory.

Ablations. We compare CAMP-II with several variants to validate its components. The variant compaction keeps its memory through the compaction protocol instead of files. The variant memory tools keeps its memory through the six dedicated tool calls instead of files. The variant gradient masked removes the loss on the tokens the policy writes into memory and keeps it on the task-action and answer tokens.

Evaluation. We report per-domain success rates and their average, with the success definitions in Appendix A. Trainable methods are trained using the CAMG-II train split and evaluated using the CAMG-II test set, and all methods use the same 128 test tasks per domain, giving 640 tasks in total. We measure external transfer on the network-free task subsets of Terminal-Bench 2.1 and SWE-Bench Pro V2 [7, 39].

6.2. Results

File-operation priors improve the task success of memory distillation. To test whether file memory is the most task-friendly form of agent memory, we compare CAMP-II against the baselines of Section 6.1 (Table 1). CAMP-II attains the highest average success rate, 64.6% versus 54.2% for CAMP-I, and the same comparison holds across five training seeds, where the five-domain mean is 64.9 with a standard deviation of 0.5 (Table 8), and only two domains reach the frozen teacher's level, Visual at 40.2% against 40.3% and Shop at 100% for both. We attribute the margin to the training signal: the teacher's token-level targets supervise each memory write as it is sampled, whereas CAMP-I must assign credit for memory operations across the long horizon, and task-reward training, as Fei et al. [10] analyze, spends much of its exploration on invalid calls and failed executions.

The memory-learning baselines fall further behind, because their channels are protocols the student must first learn, while CAMP-II's is ordinary file use. MemOPD shows what the protocol costs. Without its SFT cold start it scores zero in every domain: the every-step memory rewrite runs, but the student never submits a successful answer, and on-policy distillation cannot supervise behavior that is never sampled. With the cold start restored, MemOPD becomes the strongest memory-learning baseline. HELM shows what separation costs: its published design puts a learned memory controller in a second model, and the adaptation in which one policy fills both roles gains 13.2 points, 49.8% against 36.6%. Scaling the same recipe to a 9B student lifts the average to 66.5%, closing 63% of the student–teacher gap. Appendix C reports training dynamics for CAMP-II and CAMP-I.

The teacher and the student pay least on the file interface. To understand why training on the file interface works best, we test whether that interface lies inside what the teacher and the student already know. We measure

Method	Task success (%)					Avg.↑
	Shop	Coding	AutoResearch	General	Visual	
Frozen control						
Qwen3.5-4B	77.3	27.3	46.1	39.8	36.0	45.3
Qwen3.5-9B	90.6	26.6	56.2	60.9	36.5	54.2
Frozen teacher (Qwen3.8-27B)	100.0	63.3	83.6	93.0	40.3	76.0
Learned-memory methods						
MemOPD	0.0	0.0	0.0	0.0	0.0	0.0
MemOPD (w/ SFT)	93.0	27.3	39.1	62.5	37.3	51.8
CompactOPD	93.8	24.2	37.5	61.7	23.6	48.2
ACM	94.5	21.9	11.7	59.4	8.5	39.2
HELM	87.5	0.0	9.4	53.1	33.1	36.6
HELM (single-model)	87.5	29.7	46.9	46.9	38.0	49.8
Hy-Memory	92.2	27.3	35.9	63.3	25.8	48.9
CAMP-I	85.2	32.0	38.3	68.8	**46.5**	54.2
CAMP-II	**100.0**	**37.5**	64.1	81.2	40.2	64.6
CAMP-II-9B	**100.0**	36.7	**73.4**	**82.0**	40.2	**66.5**

Table 1. **CAMP-II attains the highest average success rate across the CAMG-II test set.** Among the 4B systems it outperforms CAMP-I, every memory-learning baseline, and the frozen student. The largest gain lands in General, and Shop saturates. Scaling the same recipe to a 9B student lifts the average further.

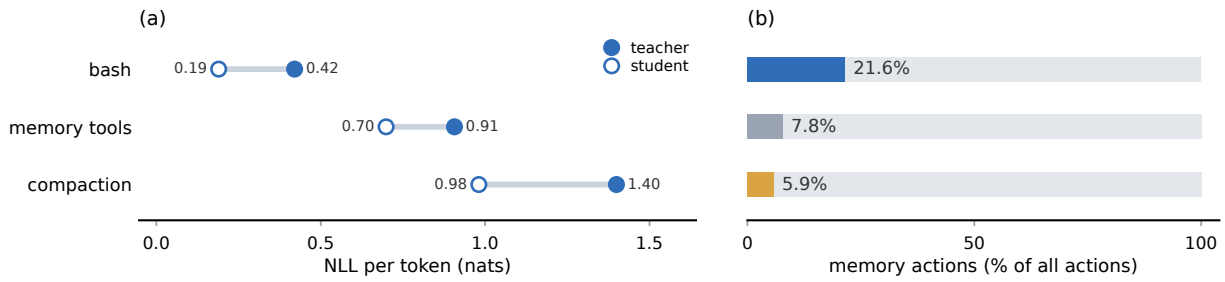


Figure 3. **Both the teacher and the student are most familiar with the file interface, and that interface carries the most memory reads and writes.** (a) Surprisal, the mean negative log-likelihood per generated token, over the five environments under the three memory interfaces. (b) Memory actions as a share of all actions. The file interface draws roughly three times the share of the dedicated tools.

how surprising the action each interface calls for is to the frozen teacher and to the initial student, the mean negative log-likelihood of the generated tokens in nats per token (Fig. 3a). The file interface costs the teacher a third of what compaction costs it, and the student a fifth, with the tool block in between. The teacher pays more at every one of them, so a stronger model does not bring these interfaces any closer. The interface also decides how finely the teacher's preference separates memory writes. Given two writes from one interface and two task contexts, the teacher ranks the write that carries the prefix's own task above the foreign one in 76.5% of decisions on the file interface, against 70.0% and 63.8% on the other two. The ordering holds when the key information of a single note is replaced by another task's (Appendix D.3).

The student reaches for memory most often on the file interface. To see whether that prior also governs what the distillation has to work with, we count the memory actions the student takes under each interface. They are 21.6% of all actions on the file interface, against 7.8% with the six dedicated tools and 5.9% under compaction (Fig. 3b). The file interface is therefore the one interface where the teacher's target is cheapest and the student already emits the action it has to be taught, which is why training there pays off. Appendix B turns that coverage into the cost of teaching the action.

The negative advantage is strongest on the unfamiliar interfaces. Once training runs, the per-token advantage can point either way. We score the memory actions the student produces under each interface and average the negative advantage, which roughly doubles from the file interface to

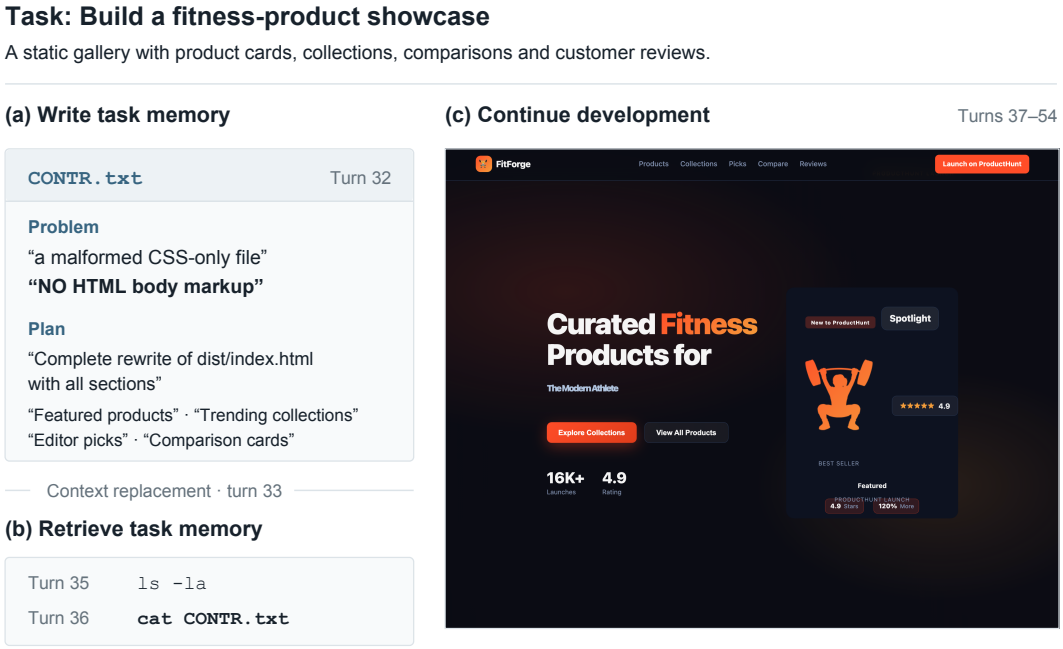


Figure 4. **A Visual episode continues from its own saved file.** We follow one episode from a late training update on the fitness-product showcase task. In (a) the student writes down what is wrong with the page and how it will rewrite it, before its context is replaced. In (b) it lists the workspace afterwards and reads that note back. In (c) it delivers a page that carries the sections the note listed.

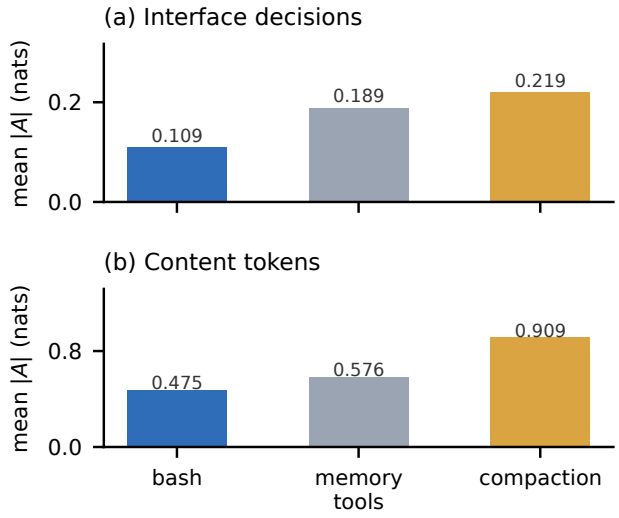


Figure 5. **The negative-advantage problem worsens as the interface becomes less familiar.** We score every token of the memory actions the student produces under each interface, with the teacher and the student. The mean size of the negative advantage rises from the file interface through the memory tools to compaction, (a) on interface decisions and (b) on content tokens.

compaction, on interface decisions and on content tokens alike (Fig. 5). The push against what the student wrote is therefore strongest where the interface is least familiar, and it supplies no target to replace what it removes. Prior work reports that a strong negative advantage destabilises on-policy distillation [16, 80].

The quality of the memory content determines task success. To test whether the quality of the memory is what carries CAMP-II's gain, we intervene on it directly, and we also ablate the supervision that reaches it. At the first context replacement we continue the same episode three ways, with the memory it wrote, with another episode's memory, or with none of it, and nothing else differs between them (Fig. 6). Success follows the content. The episode that keeps its own memory completes the sequence, the one given another episode's memory recovers part of it, and the one whose memory is emptied recovers less. When we mask the loss on the tokens the policy writes into memory and keep it on the task-action and answer tokens, the gain from distillation largely disappears (Fig. 7). Masking the same number of tokens at random instead leaves it almost intact, at 63.3 against 64.6, so the loss comes from the supervision on memory content rather than from the smaller budget. The improvement therefore lives in learning what to write into memory, which is what CAMP-II trains.

Ablating the interface. To see which part of CAMP-II carries the gain, we ablate the memory interface, replacing it with the dedicated memory tools and with compaction in

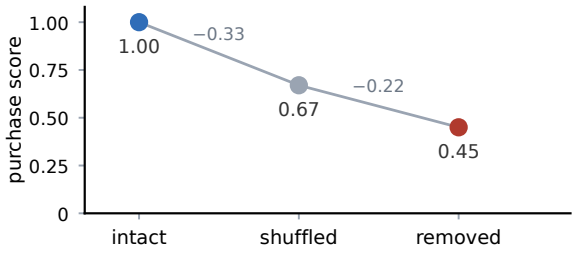


Figure 6. **An episode's later success is caused by the memory it wrote.** We continue each episode three ways at its first context replacement, leaving the file memory intact, installing another episode's memory, or emptying it, so the runs differ only in the memory that crosses the boundary.

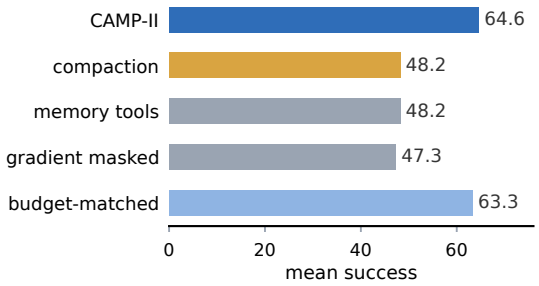


Figure 7. **Each component of CAMP-II is necessary.** On the CAMG-II test set we compare the mean success of CAMP-II with that of variants that (i) replace the file interface by compaction, (ii) replace the file interface by the dedicated memory tools, (iii) mask the loss on the memory tokens, and (iv) mask the same number of tokens at random. Every variant that touches the memory supervision loses most of the gain, and matching the masked budget does not.

turn. Either substitution costs 16 to 17 points, and both lie outside the prior of Fig. 3 (Fig. 7), so the choice of interface decides most of the gain.

Case study. We showcase the memory mechanism CAMP-II learns in a training rollout on the fitness-product showcase task. The page it has produced is a CSS-only file with no HTML body markup. Before its context is replaced, the policy records both findings and the rewrite it plans: a complete rewrite of `dist/index.html` with featured products, trending collections, editor picks and comparison cards. The replacement leaves nothing else of that reasoning. The behaviour does not depend on the requirement to write. Evaluated with the requirement removed, the trained policy scores 62.6, and retrained in that environment it scores 64.0 (Appendix D.4).

The learned memory behavior transfers outside CAMG-II. We evaluate the initial students, larger same-generation models, and the trained CAMP-II policies at 4B and 9B on Terminal-Bench 2.1 and SWE-Bench Pro V2 (Table 2). Both trained policies exceed the models they start from on both suites, the 4B policy by 6.9 points on each. Both also average above a larger model, CAMP-II-4B above the 9B model and CAMP-II-9B above the 35B one, and CAMP-II-9B holds the highest score in the table on both suites.

Table 2. **Both trained policies beat a larger same-generation model on the average, with fewer parameters.** Success rate on Terminal-Bench 2.1 and SWE-Bench Pro V2 for the initial students, same-generation larger models, and the trained policies.

Policy	TB 2.1	SWE-Bench Pro V2	Avg.
Qwen3.5-4B	23.1	11.2	17.1
Qwen3.5-9B	23.1	23.1	23.1
Qwen3.5-35B-A3B	33.8	27.4	30.6
CAMP-II-4B	30.0	18.1	24.1
CAMP-II-9B	**38.0**	**30.1**	**34.1**

7. Conclusion

We investigate how the shared pre-training priors of teachers and students affect memory-oriented on-policy distillation. Unlike task-reward RL, memory OPD requires both a student that can generate meaningful memory actions and a teacher that can provide reliable supervision over their content. To study this two-sided requirement, we introduce CAMG-II, a suite of five long-horizon agentic environments, and CAMP-II, which trains one policy across them with fully asynchronous OPD over ordinary file operations.

Our experiments show that the file interface improves both student memory-action coverage and the teacher's ability to distinguish task-relevant memory content. Memory interventions and supervision ablations further establish the role of learned memory content in downstream task performance. CAMP-II achieves 2.97 times the gain of the strongest OPD baseline on CAMG-II and generalizes to Terminal-Bench 2.1 and SWE-Bench Pro V2, where CAMP-II-9B outperforms the larger Qwen3.5-35B-A3B on both benchmarks. Shared interface priors are therefore a design consideration for memory OPD, complementary to improving distillation objectives or teacher guidance.

Limitations. The study covers students up to Qwen3.5-9B and a single frozen teacher. Whether the gains persist at larger scales or carry over to multi-teacher OPD remains untested.

LLM Usage Statement

Large language model assistants (OpenAI GPT-6.1 Sol in Codex and DeepSeek-V4.1-Flash-1017 in the DeepSeek Harness) were used to polish the writing, write experiment code, and help polish the figures and tables. All model outputs were reviewed by the authors.

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